Fiscal Year:	FY 2020	**Task Last Updated:**	FY 05/29/2020
PI Name:	Moser, Felix Ph.D.		
Project Title:	Cell-free Production of Pharmaceutical-grade Biologics		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Cambridge	**State:**	MA
Zip Code:	02139	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	TRISH--Industry
Start Date:	05/01/2019	**End Date:**	04/30/2020
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Start and end dates changed to 5/1/2019-4/30/2020 (previously 4/1/19-3/21/20) per E. Urquieta/TRISH in June 2019 (Ed., 5/29/20)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX16AO69A-IND0008		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Industry Project Biomolecules, such as proteins, carbohydrates, and metabolites, hold promise as treatments and diagnostic markers for a wide variety of diseases, including cancer, diabetes, and Alzheimer's disease. However, generating the appropriate biomolecules currently requires numerous chemicals and cell-culture systems. Synlife Inc. proposes developing freeze-dried, cell-free extracts that can generate active biomolecules upon adding only water and DNA. Further, the kit would allow purification of these biomolecules to a level safe for human injection. This system will be low cost, highly portable, and have a long shelf life.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We have shown that production of erythropoietin (EPO), granulocyte-colony-stimulating factor (G-CSF), and granulocyte macrophage colony stimulating factor (GM-CSF), three proteins that could act as treatments for acute radiation poisoning in space, could in theory be produced from freeze-dried cell-free extracts on demand. Adiga et al. (2018) showed similar results with different freeze-dried cell-free extracts and performed both the reaction and purification in automated, custom-built hardware. These proofs-of-concept show that production and purification of therapeutic proteins is possible and can be made practical through advances in method, formulation, and integrated hardware. Reference: Adiga R, Al-Adhami M, Andar A, et al. Point-of-care production of therapeutic proteins of good-manufacturing-practice quality. Nat Biomed Eng. 2018;2(9):675-686. https://
Task Progress:	Biomolecules, such as proteins, carbohydrates, and metabolites, hold promise as treatments and diagnostic markers for a wide variety of diseases, including cancer, diabetes, and Alzheimer's disease. However, generating the appropriate biomolecules currently requires numerous chemicals and cell-culture systems. Synlife, Inc. proposed developing freeze-dried, cell-free extracts that can generate active biomolecules upon adding only water and DNA. Further, their kit would allow purification of these biomolecules to a level safe for human injection. This system will be low cost, highly portable, and have a long shelf life. Multi-year spaceflight missions can encounter various unforeseen problems, not the least of which are medical. Bringing a physical pharmacopeia is impractical, so versatile tools for in-flight generation of medicines and specialty chemicals are needed. 1. Synlife envisioned building a kit containing freeze-dried cell-free extracts capable of generating active biomolecules upon addition of only DNA and water. The main objective of this proposal was to demonstrate a proof-of-concept that freeze-dried cell-free extract could rapidly generate protein therapeutics and that those proteins could be purified to pharmaceutical standards of purity. Specific objectives included: a) Demonstrate synthesis of EPO, G-CSF, GM-CSF polypeptides in cell-free extract, and b) Demonstrate purification of proteins generated in cell-free extracts. 2. We successfully completed both objectives and were able to demonstrate production and purification of EPO, G-CSF, and GM-CSF from freeze-dried PURE system. Additionally, we were able to test various conditions to optimize protein production and purification, including testing of different commercial PURE formulations. Together with published work by others (Adiga et al., 2018. Nat Biomed Eng), this work provides sufficient proof-of-concept to demonstrate that this approach has merit. 3. This project demonstrates that production and purification of human protein therapeutics from freeze-dried cell extract and recombinant systems (i.e., PURE) is possible. This raises the Technology Readiness Level (TRL) of the concept from 1 to 3 in our hands. 4. While we believe there is great merit in this technology, we have concluded the study.
Bibliography Type:	Description: (Last Updated:)